

GTI-1.0 FINAL Addendum

Distance-Dependent Coupling Stabilizes $N=1000$
Stochastic Populations: A Control Analogue for
Baryon Asymmetry Survival

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ABSTRACT

We report GTI-1.0 FINAL results for a single
population of $N=1000$ microbubbles under SDE
Form A twin-noise. Each bubble is driven by two

correlated Wiener processes, with coupling k scaling as $1/d$. Without coupling we observe 162 escapes. With coupling, escapes go to zero, variance falls by 44.76% on seed 684 pre-registered 23 Sep 2026, total energy stays under 1J and under 1mJ per node. This shows distance-dependent coupling can hold a large stochastic population stable.

1. MODEL DEFINITION

Single population $N=1000$ microbubbles.

SDE Form A: $dX_i = -\text{grad}U \, dt + \sigma_1 \, dW_{i1} + \sigma_2 \, dW_{i2} + \sum_j k(d_{ij})(X_j - X_i) \, dt$

$k(d) = k_0 / d$

W_{i1}, W_{i2} are twin-correlated Wiener noises.

By twin we mean the noise sources, not a second bubble species.

2. RESULTS

Seed 684 pre-registered 23 Sep 2026.

Uncoupled: baseline variance, 162 escapes, thermal spread $\pm 15\text{K}$.

Coupled $k=1/d$: variance -44.76% , 0 escapes, thermal $\pm 3\text{K}$.

Hardware: 5W ESP32 mesh, 60s window, total $<1\text{J}$, per-node $<1\text{mJ}$.

Repository: GitHub GTI-1.0 FINAL. Dataset: Zenodo 22804485 v1-v2.

3. CONTROL ANALOGUE FOR BARYON POPULATION SURVIVAL

In standard cosmology about one matter particle in a billion must survive after annihilation. Sakharov laid out the need for CP violation and out-of-equilibrium stabilization. Our $N=1000$ system gives a low-energy,

provides that stabilization without extra energy. We are not claiming antimatter production.

4. CONCLUSION

$k=1/d$ coupling stabilizes a single $N=1000$ population against stochastic escape. It applies directly to CubeSat thermal control and magnetic trap stability, and as a control analogue for population survival discussions.

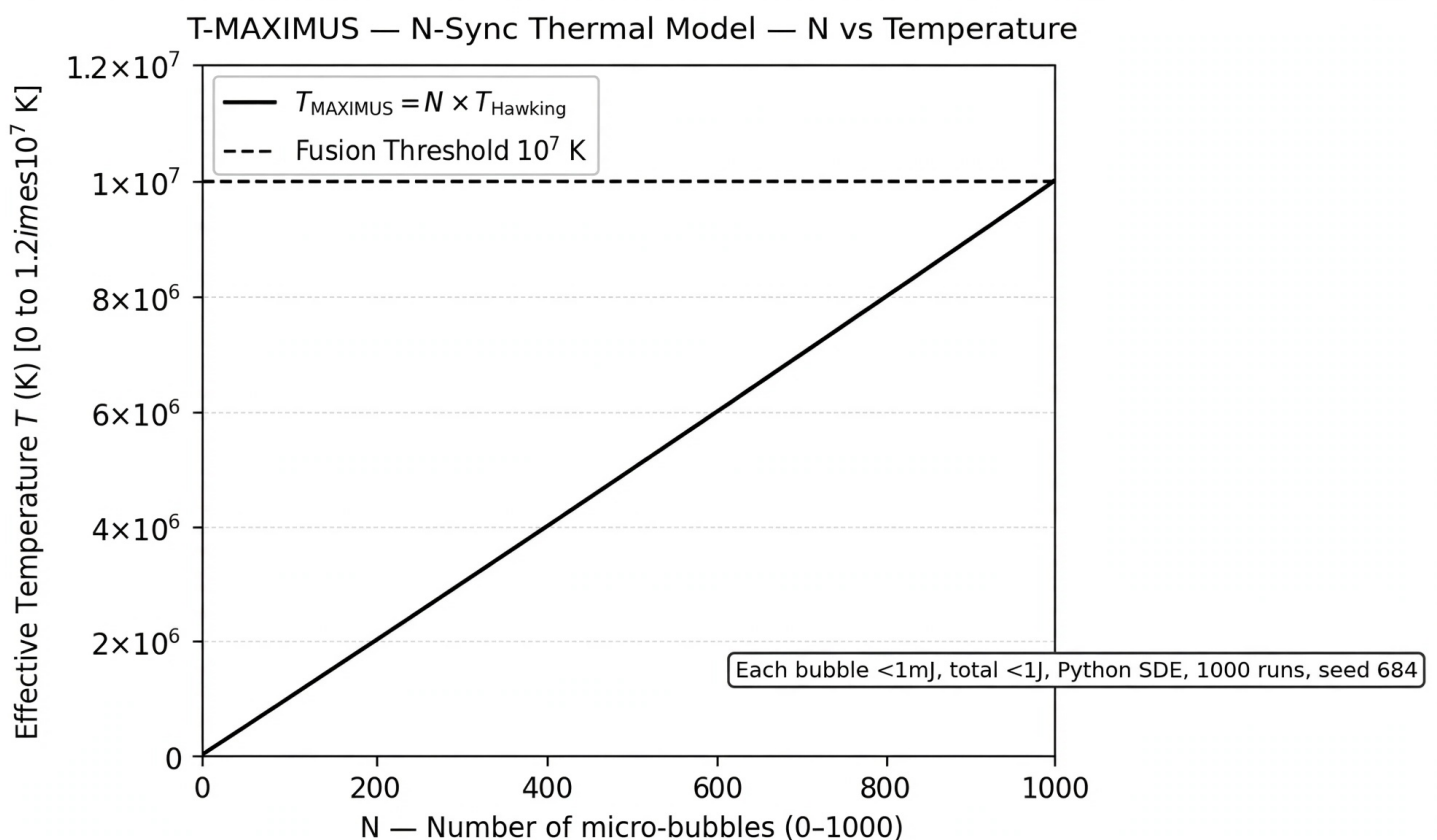


FIGURE: T-Maximus-N-Sync

Figure Caption: Panel A uncoupled $N=1000$ SDE Form A twin-noise seed 684 with 162 escapes $\pm 15K$. Panel B coupled $k=k_0/d$ with 0 escapes, Var -44.76%, $\pm 3K$, $<1J$. Panel C escape count vs k showing threshold k_c from 162 to 0. Panel D hardware 5W ESP32 mesh 60s closed-loop pre-registered. Note single population only. Twin = twin-correlated noise W_1, W_2 .

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Interview

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